COURSE REGISTRATION SYSTEM

(WINDOWS BASED APPLICATION)

Done By

INDHIRA SIVASAKTHI

INTODUCTION

A Course Registration System is a software application designed to streamline the process of enrolling in courses offered by an educational institution. This system provides a centralized platform where students can view available courses, register for classes, and manage their academic schedules. Similarly, administrators and faculty members can use the system to oversee course offerings, track student enrolment , and manage overall academic planning.

Login Form

- Users are required to enter a username and password.
- The system validates user credentials against the database using a SQL query.
- Upon successful validation, users are redirected to the Homepage.re and retrieve information related to students, courses, and feedback.

Registration Form

- Users can register with a unique username, password, email, and optional photo.
- The system checks for duplicate usernames or emails before registration.
- Upon successful registration, users receive a confirmation message.

Home Page Form

- Serves as a central navigation hub, allowing users to access different sections of the application, such as student registration, course registration, and feedback collection.

Student Registration Form

- Allows administrators to register students with various details.
- Provides validation for input fields such as name, registration number, department, semester, etc.
- Supports the optional upload of a student photo.
- Data is stored in the database upon successful registration.

Course Registration Form

- Users can input and manage course details for different semesters, including subject names, codes, theory/lab assignments, and associated staff members.
- The course information is stored in a SQL Server database.

Course Details Form

- The course details form likely a part of the Course Registration System project and is designed to display detailed information about the courses available for registration.

Feedback Collection Form

- The application collects feedback from users, including a rating based on checkboxes, technical issues, and suggestions.
- Feedback responses are stored in the database for future analysis.

Logout

- The Logout refers to the process of logging out from the application.

Application Overview

The Course Registration System is a software application designed to facilitate and optimize the course registration process within educational institutions. It aims to automate and manage various aspects of student enrolment , course selection, and registration, providing an efficient and user-friendly platform for both students and administrators.

KEY POINTS

- User Authentication
- User Registration
- Login and Registration Validation
- Course Registration
- User Profile Management
- Data Validation and Error Handling
- Registration Confirmation
- Database Integration:

USAGE

Educational Institutions: The primary users of this system are educational institutions, including schools, colleges, and universities. It streamlines the traditionally manual course registration process into a more efficient and automated workflow.

Students: Students benefit from a user-friendly interface that allows them to explore available courses, check their schedules, and register for classes with ease.

Administrators: Administrative staff can manage course offerings, track student registrations, and generate reports to analyze ,enrollment patterns.

Reduced Administrative Burden: By automating the registration process, the system reduces the administrative burden on educational institutions, allowing staff to focus on other critical tasks.

Improved Accuracy: The digital nature of the system minimizes errors associated with manual data entry and paperwork, leading to more accurate and reliable records.

SOURCE CODE

Login Form

```
using System;
using System.Collections.Generic;
using System.ComponentModel;
using System.Data;
using System.Data.SqlClient;
using System.Drawing;
using System.Linq;
using System.Text;
using System.Threading.Tasks;
using System.Windows.Forms;

namespace Project
{
    public partial class Login : Form
    {

        private const string ConnectionString = "Data
Source=LAPTOP-IUGD8B6L\\SQLEXPRESS;Initial Catalog=Project;Integrated Security=True";

        public Login()
        {
            InitializeComponent();
        }

        private void button1_Click(object sender, EventArgs e)
        {
            string username = textBox1.Text;
            string password = textBox2.Text;

            if (string.IsNullOrEmpty(username) || string.IsNullOrEmpty(password))
            {
                MessageBox.Show("Please enter both username and password.");
                return;
            }

            try
            {
                if (ValidateUser(username, password))
                {
                    OpenHomeForm();
                }
                else
                {
                    MessageBox.Show("Invalid username or password. Please try again.");
                }
            }
            catch (Exception ex)
            {
                MessageBox.Show($"An error occurred: {ex.Message}");
            }
        }
        private bool ValidateUser(string username, string password)
```

```
        {
            using (SqlConnection connection = new SqlConnection(ConnectionString))
            {
                connection.Open();
                string query = "SELECT COUNT(1) FROM Users WHERE Username = @Username AND 
Password = @Password";

                using (SqlCommand command = new SqlCommand(query, connection))
                {
                    command.Parameters.AddWithValue("@Username", username);
                    command.Parameters.AddWithValue("@Password", password);

                    int count = (int)command.ExecuteScalar();
                    return count == 1;
                }
            }
        }



        private void OpenHomeForm()
        {
            Homepage homePage = new Homepage();
            homePage.Show();
            this.Hide();
        }
        private void OpenRegisterForm()
        {
            Register registerForm = new Register();
            registerForm.Show();

        }

        private void Form1_Load(object sender, EventArgs e)
        {

        }

        private void button2_Click(object sender, EventArgs e)
        {
            OpenRegisterForm();
        }
        private void label2_Click(object sender, EventArgs e)
        {

        }

    }
}
```

Register Form

```
using System;
using System.Collections.Generic;
using System.ComponentModel;
using System.Data;
using System.Data.SqlClient;
using System.Drawing;
using System.Linq;
using System.Text;
using System.Threading.Tasks;
using System.Windows.Forms;

namespace Project
{
    public partial class Register : Form
    {
        private const string ConnectionString = "Data
Source=LAPTOP-IUGD8B6L\\SQLEXPRESS;Initial Catalog=Project;Integrated Security=True";

        public Register()
        {
            InitializeComponent();
        }

        private void button1_Click(object sender, EventArgs e)
        {
            string username = textBox1.Text;
            string password = textBox2.Text;
            string confirmPassword = textBox3.Text;
            string email = textBox4.Text;
            if (string.IsNullOrEmpty(username) || string.IsNullOrEmpty(password) ||
string.IsNullOrEmpty(confirmPassword) || string.IsNullOrEmpty(email))
            {
                MessageBox.Show("Please fill in all fields.");
                return;
            }
            if (password != confirmPassword)
            {
                MessageBox.Show("Passwords do not match. Please try again.");
                return;
            }

            try
            {
                if (IsUsernameOrEmailTaken(username, email))
                {
                    MessageBox.Show("Username or email is already taken. Please choose different ones.");
                    return;
                }
                if (RegisterUser(username, password, email))
                {
                    MessageBox.Show("Registration successful!");
                    this.DialogResult = DialogResult.OK;
                    this.Close();
```

```
            }
            else
            {
                MessageBox.Show("Error occurred during registration. Please try again.");
            }
        }
        catch (Exception ex)
        {

            MessageBox.Show($"An error occurred: {ex.Message}");
        }
    }

    private bool IsUsernameOrEmailTaken(string username, string email)
    {
        using (SqlConnection connection = new SqlConnection(ConnectionString))
        {
            connection.Open();


   string query = "SELECT COUNT(1) FROM Users WHERE Username = @Username OR Email
= @Email";

            using (SqlCommand command = new SqlCommand(query, connection))
            {
                command.Parameters.AddWithValue("@Username", username);
                command.Parameters.AddWithValue("@Email", email);
                int count = (int)command.ExecuteScalar();
                return count > 0;
            }
        }
    }
    private void label2_Click(object sender, EventArgs e)
    {

    }
    private bool RegisterUser(string username, string password, string email)
    {
        using (SqlConnection connection = new SqlConnection(ConnectionString))
        {
            connection.Open();

            string query = "INSERT INTO Users (Username, Password, Email) VALUES (@Username,
@Password, @Email)";

            using (SqlCommand command = new SqlCommand(query, connection))
            {
                command.Parameters.AddWithValue("@Username", username);
                command.Parameters.AddWithValue("@Password", password);
                command.Parameters.AddWithValue("@Email", email);

                int rowsAffected = command.ExecuteNonQuery();

                return rowsAffected > 0;
            }
```

```
            }

        }


    }
}
```

Home Page :

```
using System;
using System.Collections.Generic;
using System.ComponentModel;
using System.Data;
using System.Drawing;
using System.Linq;
using System.Text;
using System.Threading.Tasks;
using System.Windows.Forms;

namespace Project
{
    public partial class Homepage : Form
    {
        public Homepage()
        {
            InitializeComponent();
        }

        private void button5_Click(object sender, EventArgs e)
        {
            this.Close();
        }

        private void button1_Click(object sender, EventArgs e)
        {
            OpenStudentForm();
        }
        private void OpenStudentForm()
        {
            StudentRegistrationForm Studentform = new StudentRegistrationForm();
            Studentform.Show();
            this.Hide();
        }

        private void button2_Click(object sender, EventArgs e)
        {
            OpenCourseForm();

        }
        private void OpenCourseForm()
        {
            CourseRegistrationForm Courseform = new CourseRegistrationForm();
            Courseform.Show();
```

```
            this.Hide();
        }

        private void button3_Click(object sender, EventArgs e)
        {
            OpenFeedbackForm();
        }
        private void OpenFeedbackForm()
        {
            FeedbackForm feedbackform = new FeedbackForm();
            feedbackform.Show();
            this.Hide();
        }

        private void label1_Click(object sender, EventArgs e)
        {

        }
    }
}
```

Student Registration Form:

```
using System;
using System.Collections.Generic;
using System.ComponentModel;
using System.Data;
using System.Data.SqlClient;
using System.Drawing;
using System.Linq;
using System.Text;
using System.Threading.Tasks;
using System.Windows.Forms;
using System.IO;

namespace Project
{
    public partial class StudentRegistrationForm : Form
    {
        private const string ConnectionString = "Data 
Source=LAPTOP-IUGD8B6L\\SQLEXPRESS;Initial Catalog=Project;Integrated Security=True";

        public StudentRegistrationForm()
        {
            InitializeComponent();


        }

        private void button2_Click(object sender, EventArgs e)
        {
            if (ValidateInput())
            {
                SaveDataToDatabase();
```

```
            }
        }
        private bool ValidateInput()
        {
            if (string.IsNullOrWhiteSpace(textBox1.Text))
            {
                MessageBox.Show("Please enter a valid name.", "Validation Error",
MessageBoxButtons.OK, MessageBoxIcon.Error);
                return false;
            }
            if (string.IsNullOrWhiteSpace(textBox2.Text))
            {
                MessageBox.Show("Please enter a valid register number.", "Validation Error",
MessageBoxButtons.OK, MessageBoxIcon.Error);
                return false;
            }

            if (comboBox1.SelectedIndex == -1)
            {
                MessageBox.Show("Please select a department.", "Validation Error",
MessageBoxButtons.OK, MessageBoxIcon.Error);
                return false;
            }
            if (comboBox3.SelectedIndex == -1)
            {
                MessageBox.Show("Please select a semester.", "Validation Error",
MessageBoxButtons.OK, MessageBoxIcon.Error);
                return false;
            }

            if (comboBox2.SelectedIndex == -1)
            {
                MessageBox.Show("Please select a year.", "Validation Error", MessageBoxButtons.OK,
MessageBoxIcon.Error);
                return false;
            }

            if (!IsValidEmail(textBox3.Text))
            {
                MessageBox.Show("Please enter a valid email address.", "Validation Error",
MessageBoxButtons.OK, MessageBoxIcon.Error);
                return false;
            }
            if (comboBox4.SelectedIndex == -1)
            {
                MessageBox.Show("Please select a Programme.", "Validation Error",
MessageBoxButtons.OK, MessageBoxIcon.Error);
                return false;
            }
            if (string.IsNullOrWhiteSpace(textBox4.Text))
            {
                MessageBox.Show("Please enter a valid contact number.", "Validation Error",
MessageBoxButtons.OK, MessageBoxIcon.Error);
                return false;
            }
```

```
            if (string.IsNullOrWhiteSpace(textBox5.Text))
            {
                MessageBox.Show("Please enter a valid section.", "Validation Error",
MessageBoxButtons.OK, MessageBoxIcon.Error);
                return false;
            }
            if (string.IsNullOrWhiteSpace(textBox6.Text))
            {
                MessageBox.Show("Please enter a valid regulation.", "Validation Error",
MessageBoxButtons.OK, MessageBoxIcon.Error);
                return false;
            }
            return true;
        }

        private bool IsValidEmail(string email)
        {

            return System.Text.RegularExpressions.Regex.IsMatch(email,
                @"^[a-zA-Z0-9_.+-]+@[a-zA-Z0-9-]+\.[a-zA-Z0-9-.]+$");
        }
        private void button1_Click(object sender, EventArgs e)
        {

            textBox1.Text = "";
            textBox2.Text = "";
            textBox3.Text = "";
            textBox4.Text = "";
            textBox5.Text = "";
            textBox6.Text = "";
            comboBox1.SelectedIndex = -1;
            comboBox2.SelectedIndex = -1;
            comboBox3.SelectedIndex = -1;
            comboBox4.SelectedIndex = -1;
            pictureBox1.Image = null;
        }

        private void SaveDataToDatabase()
        {
            using (SqlConnection connection = new SqlConnection(ConnectionString))
            {
                connection.Open();

                string query = "INSERT INTO Students (Name, RegisterNo, Department, Semester, Year,
Email,ContactNo,Section,Regulation,Programme,Photo) " +
                               "VALUES (@Name, @RegisterNo, @Department, @Semester, @Year,
@Email,@ContactNo,@Section,@Regulation,@Programme, @Photo)";

                using (SqlCommand command = new SqlCommand(query, connection))
                {
                    command.Parameters.AddWithValue("@Name", textBox1.Text);
                    command.Parameters.AddWithValue("@RegisterNo", textBox2.Text);
                    command.Parameters.AddWithValue("@Department",
comboBox1.SelectedItem.ToString());
```

```
                command.Parameters.AddWithValue("@Semester",
comboBox3.SelectedItem.ToString());
                command.Parameters.AddWithValue("@Year", comboBox2.SelectedItem.ToString());
                command.Parameters.AddWithValue("@Email", textBox3.Text);
                command.Parameters.AddWithValue("@ContactNo", textBox4.Text);
                command.Parameters.AddWithValue("@Section", textBox5.Text);
                command.Parameters.AddWithValue("@Regulation", textBox6.Text);
                command.Parameters.AddWithValue("@Programme",
comboBox4.SelectedItem.ToString());


                if (pictureBox1.Image != null)
                {

                    byte[] photoBytes = ImageToByteArray(pictureBox1.Image);
                    command.Parameters.AddWithValue("@Photo", photoBytes);
                }
                else
                {
                    command.Parameters.AddWithValue("@Photo", DBNull.Value);
                }
                int rowsAffected = command.ExecuteNonQuery();

                if (rowsAffected > 0)
                {
                    MessageBox.Show("Student registration successful!");
                }
                else
                {
                    MessageBox.Show("Error occurred during student registration. Please try again.");
                }
            }
        }
    }

    private byte[] ImageToByteArray(Image image)
    {
        using (MemoryStream ms = new MemoryStream())
        {
            image.Save(ms, System.Drawing.Imaging.ImageFormat.Jpeg);
            return ms.ToArray();
        }
    }

    private void button3_Click(object sender, EventArgs e)
    {
        using (OpenFileDialog openFileDialog = new OpenFileDialog())
        {
            openFileDialog.Filter = "Image Files|*.jpg;*.jpeg;*.png;*.gif|All Files|*.*";

            if (openFileDialog.ShowDialog() == DialogResult.OK)
            {
                pictureBox1.Image = new System.Drawing.Bitmap(openFileDialog.FileName);
            }
        }
```

```
        }


        private void button4_Click(object sender, EventArgs e)
        {
            OpenHomeForm();
        }


        private void OpenHomeForm()
        {
            Homepage homePage = new Homepage();
            homePage.Show();
            this.Hide(); // Hide the current form
        }

    }
}
```

Course Registration Form

```
using System;
using System.Collections.Generic;
using System.ComponentModel;
using System.Data;
using System.Drawing;
using System.Linq;
using System.Text;
using System.Threading.Tasks;
using System.Windows.Forms;
using System.Data.SqlClient;
using System.IO;

namespace Project
{
    public partial class CourseRegistrationForm : Form
    {
        private const string ConnectionString = "Data
Source=LAPTOP-IUGD8B6L\\SQLEXPRESS;Initial Catalog=Project;Integrated Security=True";

        public CourseRegistrationForm()
        {
            InitializeComponent();
        }

        private void CourseRegistrationForm_Load(object sender, EventArgs e)
        {

        }

        private void button1_Click(object sender, EventArgs e)
        {

            if (string.IsNullOrWhiteSpace(textBox1.Text))
            {
```

```
            MessageBox.Show("Please enter a valid registration number.", "Validation Error",
MessageBoxButtons.OK, MessageBoxIcon.Error);
            return;
        }

        FetchStudentDetails(textBox1.Text);
    }
    private void FetchStudentDetails(string registrationNumber)
    {
        using (SqlConnection connection = new SqlConnection(ConnectionString))
        {
            connection.Open();

            string query = "SELECT Name, Programme, Department, Semester, Photo FROM Students
WHERE RegisterNo = @RegisterNo";

            using (SqlCommand command = new SqlCommand(query, connection))
            {
                command.Parameters.AddWithValue("@RegisterNo", registrationNumber);

                using (SqlDataReader reader = command.ExecuteReader())
                {
                    if (reader.Read())
                    {
                        label7.Text = "NAME : " + reader["Name"].ToString();
                        label8.Text = "PROGRAMME : " + reader["Programme"].ToString();
                        label9.Text = "DEPARTMENT : " + reader["Department"].ToString();
                        label10.Text = "SEMESTER :" + reader["Semester"].ToString();

                        if (reader["Photo"] != DBNull.Value)
                        {
                            byte[] photoBytes = (byte[])reader["Photo"];
                            pictureBox1.Image = ByteArrayToImage(photoBytes);
                        }
                        else
                        {
                            pictureBox1.Image = null;
                        }
                    }
                    else
                    {
                        MessageBox.Show("Student not found with the provided registration number.",
"Error", MessageBoxButtons.OK, MessageBoxIcon.Error);
                    }
                }
            }
        }
    }

    private Image ByteArrayToImage(byte[] byteArray)
    {
        using (MemoryStream ms = new MemoryStream(byteArray))
        {
            return Image.FromStream(ms);
        }
```

```
        }

        private void button2_Click(object sender, EventArgs e)
        {
            OpenCourseDetails();
        }
        private void OpenCourseDetails()
        {
            CourseDetailsForm semesterone = new CourseDetailsForm();
            semesterone.Show();
            this.Hide();
        }

        private void label2_Click(object sender, EventArgs e)
        {

        }

        private void label3_Click(object sender, EventArgs e)
        {

        }
    }
}
```

Course Details Form:

```
using System;
using System.Data.SqlClient;
using System.Windows.Forms;

namespace Project
{
    public partial class CourseDetailsForm : Form
    {
        private const string ConnectionString = "Data
Source=LAPTOP-IUGD8B6L\\SQLEXPRESS;Initial Catalog=Project;Integrated Security=True";

        public CourseDetailsForm()
        {
            InitializeComponent();
        }

        private void CourseDetailsForm_Load(object sender, EventArgs e)
        {

        }

        private void button2_Click(object sender, EventArgs e)
        {
            try
            {

                if (!AreAllFieldsFilled())
                {
```

```
                MessageBox.Show("Please fill in all fields.");
                return;
            }


            SaveDataToDatabase();
            AreAllFieldsFilled();
            ClearFormData();

        }
        catch (Exception ex)
        {
            MessageBox.Show($"An error occurred: {ex.Message}");
        }
    }

    private void SaveDataToDatabase()
    {
        try
        {
            using (SqlConnection connection = new SqlConnection(ConnectionString))
            {
                connection.Open();

                string insertQuery = "INSERT INTO CourseDetails (RegisterNo, Semester, Subject1,
SubjectCode1, TheoryLab1, Staff1, Subject2, SubjectCode2, TheoryLab2, Staff2, Subject3,
SubjectCode3, TheoryLab3, Staff3, Subject4, SubjectCode4, TheoryLab4, Staff4, Subject5,
SubjectCode5, TheoryLab5, Staff5, Subject6, SubjectCode6, TheoryLab6, Staff6, Subject7,
SubjectCode7, TheoryLab7, Staff7) " +
                                     "VALUES (@RegisterNo, @Semester, @Subject1, @SubjectCode1,
@TheoryLab1, @Staff1, @Subject2, @SubjectCode2, @TheoryLab2, @Staff2, @Subject3,
@SubjectCode3, @TheoryLab3, @Staff3, @Subject4, @SubjectCode4, @TheoryLab4, @Staff4,
@Subject5, @SubjectCode5, @TheoryLab5, @Staff5, @Subject6, @SubjectCode6, @TheoryLab6,
@Staff6, @Subject7, @SubjectCode7, @TheoryLab7, @Staff7)";

                using (SqlCommand command = new SqlCommand(insertQuery, connection))
                {
                    command.Parameters.AddWithValue("@RegisterNo", textBox1.Text);
                command.Parameters.AddWithValue("@Semester", textBox9.Text);
                command.Parameters.AddWithValue("@Subject1", comboBox1.Text);
                command.Parameters.AddWithValue("@SubjectCode1", textBox2.Text);
                command.Parameters.AddWithValue("@TheoryLab1", comboBox29.Text);
                command.Parameters.AddWithValue("@Staff1", comboBox36.Text);


                command.Parameters.AddWithValue("@Subject2", comboBox2.Text);
                command.Parameters.AddWithValue("@SubjectCode2", textBox3.Text);
                command.Parameters.AddWithValue("@TheoryLab2", comboBox30.Text);
                command.Parameters.AddWithValue("@Staff2", comboBox37.Text);

                command.Parameters.AddWithValue("@Subject3", comboBox3.Text);
                command.Parameters.AddWithValue("@SubjectCode3", textBox4.Text);
                command.Parameters.AddWithValue("@TheoryLab3", comboBox31.Text);
                command.Parameters.AddWithValue("@Staff3", comboBox38.Text);
```

```
            command.Parameters.AddWithValue("@Subject4", comboBox4.Text);
            command.Parameters.AddWithValue("@SubjectCode4", textBox5.Text);
            command.Parameters.AddWithValue("@TheoryLab4", comboBox32.Text);
            command.Parameters.AddWithValue("@Staff4", comboBox39.Text);


            command.Parameters.AddWithValue("@Subject5", comboBox5.Text);
            command.Parameters.AddWithValue("@SubjectCode5", textBox6.Text);
            command.Parameters.AddWithValue("@TheoryLab5", comboBox33.Text);
            command.Parameters.AddWithValue("@Staff5", comboBox40.Text);


            command.Parameters.AddWithValue("@Subject6", comboBox6.Text);
            command.Parameters.AddWithValue("@SubjectCode6", textBox7.Text);
            command.Parameters.AddWithValue("@TheoryLab6", comboBox34.Text);
            command.Parameters.AddWithValue("@Staff6", comboBox41.Text);


            command.Parameters.AddWithValue("@Subject7", comboBox7.Text);
            command.Parameters.AddWithValue("@SubjectCode7", textBox8.Text);
            command.Parameters.AddWithValue("@TheoryLab7", comboBox35.Text);
            command.Parameters.AddWithValue("@Staff7", comboBox42.Text);

              int rowsAffected = command.ExecuteNonQuery();

              if (rowsAffected > 0)
              {
                MessageBox.Show("Course Registration successful!");

              }
              else
              {
                MessageBox.Show("Registration failed. Please try again.");
              }
              OpenHomeForm();
            }
          }
        }
        catch (Exception ex)
        {
          MessageBox.Show($"An error occurred: {ex.Message}");
        }

      }

      private bool AreAllFieldsFilled()
      {

        return !string.IsNullOrWhiteSpace(textBox1.Text) &&
            !string.IsNullOrWhiteSpace(textBox9.Text) &&
            !string.IsNullOrWhiteSpace(comboBox1.Text) &&
            !string.IsNullOrWhiteSpace(textBox2.Text) &&
            !string.IsNullOrWhiteSpace(comboBox29.Text) &&
            !string.IsNullOrWhiteSpace(comboBox36.Text) &&
```

```
        !string.IsNullOrWhiteSpace(comboBox2.Text) &&
        !string.IsNullOrWhiteSpace(textBox3.Text) &&
        !string.IsNullOrWhiteSpace(comboBox30.Text) &&
        !string.IsNullOrWhiteSpace(comboBox37.Text) &&

        !string.IsNullOrWhiteSpace(comboBox3.Text) &&
        !string.IsNullOrWhiteSpace(textBox4.Text) &&
        !string.IsNullOrWhiteSpace(comboBox31.Text) &&
        !string.IsNullOrWhiteSpace(comboBox38.Text) &&

        !string.IsNullOrWhiteSpace(comboBox4.Text) &&
        !string.IsNullOrWhiteSpace(textBox5.Text) &&
        !string.IsNullOrWhiteSpace(comboBox32.Text) &&
        !string.IsNullOrWhiteSpace(comboBox39.Text) &&

        !string.IsNullOrWhiteSpace(comboBox5.Text) &&
        !string.IsNullOrWhiteSpace(textBox6.Text) &&
        !string.IsNullOrWhiteSpace(comboBox33.Text) &&
        !string.IsNullOrWhiteSpace(comboBox40.Text) &&

        !string.IsNullOrWhiteSpace(comboBox6.Text) &&
        !string.IsNullOrWhiteSpace(textBox7.Text) &&
        !string.IsNullOrWhiteSpace(comboBox34.Text) &&
        !string.IsNullOrWhiteSpace(comboBox41.Text) &&

        !string.IsNullOrWhiteSpace(comboBox7.Text) &&
        !string.IsNullOrWhiteSpace(textBox8.Text) &&
        !string.IsNullOrWhiteSpace(comboBox35.Text) &&
        !string.IsNullOrWhiteSpace(comboBox42.Text);


}
private void ClearFormData()
{

    textBox1.Text = string.Empty;
    textBox9.Text = string.Empty;

    comboBox1.SelectedIndex = -1;
    textBox2.Text = string.Empty;
    comboBox29.SelectedIndex = -1;
    comboBox36.SelectedIndex = -1;



    comboBox2.SelectedIndex = -1;
    textBox3.Text = string.Empty;
    comboBox30.SelectedIndex = -1;
    comboBox37.SelectedIndex = -1;

    comboBox3.SelectedIndex = -1;
    textBox4.Text = string.Empty;
    comboBox31.SelectedIndex = -1;
    comboBox38.SelectedIndex = -1;
```

```
            comboBox4.SelectedIndex = -1;
            textBox5.Text = string.Empty;
            comboBox32.SelectedIndex = -1;
            comboBox39.SelectedIndex = -1;

            comboBox5.SelectedIndex = -1;
            textBox6.Text = string.Empty;
            comboBox33.SelectedIndex = -1;
            comboBox40.SelectedIndex = -1;

            comboBox6.SelectedIndex = -1;
            textBox7.Text = string.Empty;
            comboBox34.SelectedIndex = -1;
            comboBox41.SelectedIndex = -1;

            comboBox7.SelectedIndex = -1;
            textBox8.Text = string.Empty;
            comboBox35.SelectedIndex = -1;
            comboBox42.SelectedIndex = -1;


        }


        private void OpenHomeForm()
        {
            Homepage homepage = new Homepage();
            homepage.Show();
            this.Hide();
        }

        private void button1_Click(object sender, EventArgs e)
        {
            OpenHomeForm();
        }

        private void button3_Click(object sender, EventArgs e)
        {
            string registerNo = textBox1.Text.Trim();

            if (!string.IsNullOrEmpty(registerNo))
            {
                FetchAndDisplayCourseDetails(registerNo);
            }
            else
            {
                MessageBox.Show("Please enter a valid register number.", "Validation Error",
MessageBoxButtons.OK, MessageBoxIcon.Error);
            }
        }

        private void FetchAndDisplayCourseDetails(string registerNo)
        {
            try
```

```
{
  using (SqlConnection connection = new SqlConnection(ConnectionString))
  {
    connection.Open();

    string query = "SELECT Semester, Subject1, SubjectCode1, TheoryLab1, Staff1, " +
            "Subject2, SubjectCode2, TheoryLab2, Staff2, " +
            "Subject3, SubjectCode3, TheoryLab3, Staff3, " +
            "Subject4, SubjectCode4, TheoryLab4, Staff4, " +
            "Subject5, SubjectCode5, TheoryLab5, Staff5, " +
            "Subject6, SubjectCode6, TheoryLab6, Staff6, " +
            "Subject7, SubjectCode7, TheoryLab7, Staff7 " +
            "FROM CourseDetails WHERE RegisterNo = @RegisterNo";

    using (SqlCommand command = new SqlCommand(query, connection))
    {
      command.Parameters.AddWithValue("@RegisterNo", registerNo);

      using (SqlDataReader reader = command.ExecuteReader())
      {
        if (reader.Read())
        {

          textBox9.Text = reader["Semester"].ToString();
          comboBox1.Text = reader["Subject1"].ToString();
          textBox2.Text = reader["SubjectCode1"].ToString();
          comboBox29.Text = reader["TheoryLab1"].ToString();
          comboBox36.Text = reader["Staff1"].ToString();

          comboBox2.Text = reader["Subject2"].ToString();
          textBox3.Text = reader["SubjectCode2"].ToString();
          comboBox30.Text = reader["TheoryLab2"].ToString();
          comboBox37.Text = reader["Staff2"].ToString();

          comboBox3.Text = reader["Subject3"].ToString();
          textBox4.Text = reader["SubjectCode3"].ToString();
          comboBox31.Text = reader["TheoryLab3"].ToString();
          comboBox38.Text = reader["Staff3"].ToString();

          comboBox4.Text = reader["Subject4"].ToString();
          textBox5.Text = reader["SubjectCode4"].ToString();
          comboBox32.Text = reader["TheoryLab4"].ToString();
          comboBox39.Text = reader["Staff4"].ToString();

          comboBox5.Text = reader["Subject5"].ToString();
          textBox6.Text = reader["SubjectCode5"].ToString();
          comboBox33.Text = reader["TheoryLab5"].ToString();
          comboBox40.Text = reader["Staff5"].ToString();

          comboBox6.Text = reader["Subject6"].ToString();
          textBox7.Text = reader["SubjectCode6"].ToString();
          comboBox34.Text = reader["TheoryLab6"].ToString();
          comboBox41.Text = reader["Staff6"].ToString();

          comboBox7.Text = reader["Subject7"].ToString();
```

```
                        textBox8.Text = reader["SubjectCode7"].ToString();
                        comboBox35.Text = reader["TheoryLab7"].ToString();
                        comboBox42.Text = reader["Staff7"].ToString();
                    }
                    else
                    {
                        MessageBox.Show("Course details not found with the provided register number.",
"Error", MessageBoxButtons.OK, MessageBoxIcon.Error);
                    }
                }
            }
        }
    }
    catch (Exception ex)
    {
        MessageBox.Show($"An error occurred: {ex.Message}");
    }
}

private void button4_Click(object sender, EventArgs e)
{
    if (AreAllFieldsFilled())
    {
        UpdateCourseDetailsInDatabase();
    }
    else
    {
        MessageBox.Show("Please fill in all fields.");
    }
}

private void UpdateCourseDetailsInDatabase()
{
    try
    {
        using (SqlConnection connection = new SqlConnection(ConnectionString))
        {
            connection.Open();

            string updateQuery = "UPDATE CourseDetails SET " +
                                 "Semester = @Semester, " +
                                 "Subject1 = @Subject1, SubjectCode1 = @SubjectCode1, TheoryLab1 =
@TheoryLab1, Staff1 = @Staff1, " +
                                 "Subject2 = @Subject2, SubjectCode2 = @SubjectCode2, TheoryLab2 =
@TheoryLab2, Staff2 = @Staff2, " +
                                 "Subject3 = @Subject3, SubjectCode3 = @SubjectCode3, TheoryLab3 =
@TheoryLab3, Staff3 = @Staff3, " +
                                 "Subject4 = @Subject4, SubjectCode4 = @SubjectCode4, TheoryLab4 =
@TheoryLab4, Staff4 = @Staff4, " +
                                 "Subject5 = @Subject5, SubjectCode5 = @SubjectCode5, TheoryLab5 =
@TheoryLab5, Staff5 = @Staff5, " +
                                 "Subject6 = @Subject6, SubjectCode6 = @SubjectCode6, TheoryLab6 =
@TheoryLab6, Staff6 = @Staff6, " +
                                 "Subject7 = @Subject7, SubjectCode7 = @SubjectCode7, TheoryLab7 =
@TheoryLab7, Staff7 = @Staff7 " +
```

```
                    "WHERE RegisterNo = @RegisterNo";

            using (SqlCommand command = new SqlCommand(updateQuery, connection))
            {
                SetCommandParameters(command);

                int rowsAffected = command.ExecuteNonQuery();

                if (rowsAffected > 0)
                {
                    MessageBox.Show("Course details updated successfully!");
                }
                else
                {
                    MessageBox.Show("Update failed. Please try again.");
                }
            }
        }
    }
    catch (Exception ex)
    {
        MessageBox.Show($"An error occurred: {ex.Message}");
    }
}

private void SetCommandParameters(SqlCommand command)
{
    command.Parameters.AddWithValue("@RegisterNo", textBox1.Text);
    command.Parameters.AddWithValue("@Semester", textBox9.Text);
    command.Parameters.AddWithValue("@Subject1", comboBox1.Text);
    command.Parameters.AddWithValue("@SubjectCode1", textBox2.Text);
    command.Parameters.AddWithValue("@TheoryLab1", comboBox29.Text);
    command.Parameters.AddWithValue("@Staff1", comboBox36.Text);

    command.Parameters.AddWithValue("@Subject2", comboBox2.Text);
    command.Parameters.AddWithValue("@SubjectCode2", textBox3.Text);
    command.Parameters.AddWithValue("@TheoryLab2", comboBox30.Text);
    command.Parameters.AddWithValue("@Staff2", comboBox37.Text);

    command.Parameters.AddWithValue("@Subject3", comboBox3.Text);
    command.Parameters.AddWithValue("@SubjectCode3", textBox4.Text);
    command.Parameters.AddWithValue("@TheoryLab3", comboBox31.Text);
    command.Parameters.AddWithValue("@Staff3", comboBox38.Text);

    command.Parameters.AddWithValue("@Subject4", comboBox4.Text);
    command.Parameters.AddWithValue("@SubjectCode4", textBox5.Text);
    command.Parameters.AddWithValue("@TheoryLab4", comboBox32.Text);
    command.Parameters.AddWithValue("@Staff4", comboBox39.Text);

    command.Parameters.AddWithValue("@Subject5", comboBox5.Text);
    command.Parameters.AddWithValue("@SubjectCode5", textBox6.Text);
    command.Parameters.AddWithValue("@TheoryLab5", comboBox33.Text);
    command.Parameters.AddWithValue("@Staff5", comboBox40.Text);

    command.Parameters.AddWithValue("@Subject6", comboBox6.Text);
```

```
            command.Parameters.AddWithValue("@SubjectCode6", textBox7.Text);
            command.Parameters.AddWithValue("@TheoryLab6", comboBox34.Text);
            command.Parameters.AddWithValue("@Staff6", comboBox41.Text);

            command.Parameters.AddWithValue("@Subject7", comboBox7.Text);
            command.Parameters.AddWithValue("@SubjectCode7", textBox8.Text);
            command.Parameters.AddWithValue("@TheoryLab7", comboBox35.Text);
            command.Parameters.AddWithValue("@Staff7", comboBox42.Text);
        }

    }
}
```

Feedback Form :

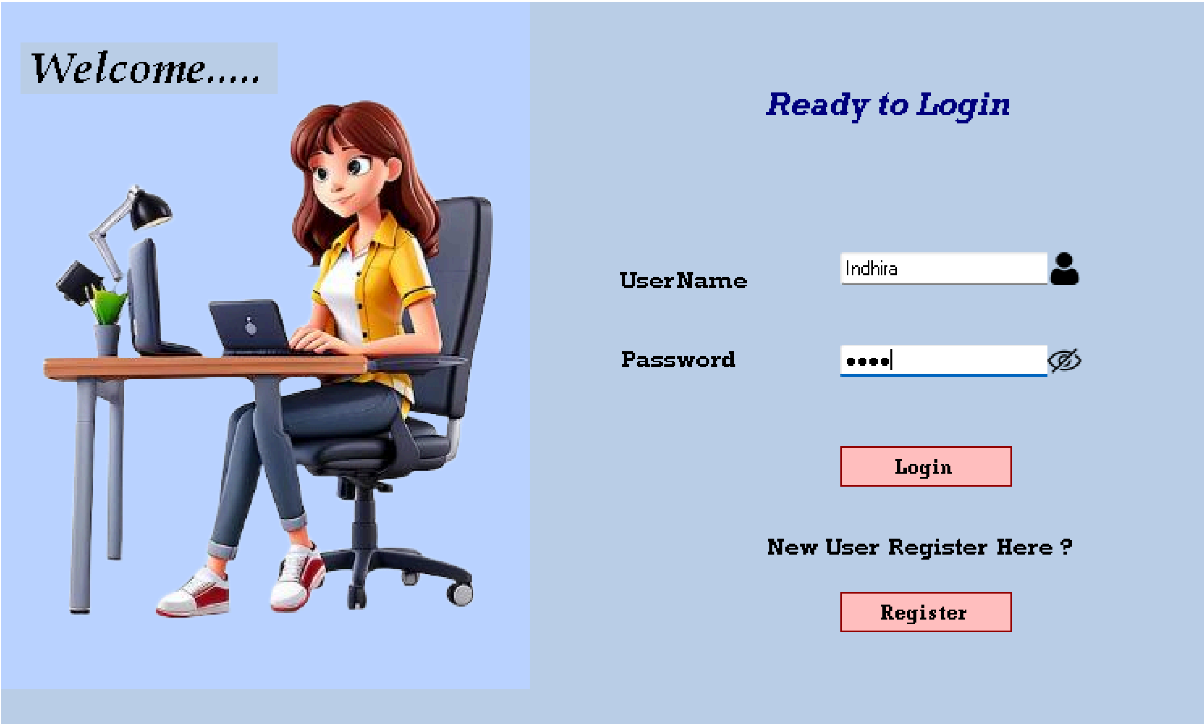
```
using System;
using System.Collections.Generic;
using System.ComponentModel;
using System.Data;
using System.Drawing;
using System.Linq;
using System.Text;
using System.Threading.Tasks;
using System.Windows.Forms;
using System.Data.SqlClient;

namespace Project
{
    public partial class FeedbackForm : Form
    {
        private const string ConnectionString = "Data
Source=LAPTOP-IUGD8B6L\\SQLEXPRESS;Initial Catalog=Project;Integrated Security=True";

        public FeedbackForm()
        {
            InitializeComponent();
        }

        private void button1_Click(object sender, EventArgs e)
        {

            int easeRating = GetCheckBoxRating(groupBox1);
            string technicalIssue = textBox1.Text;
            string suggestion = textBox2.Text;


            SaveFeedbackToDatabase(easeRating, technicalIssue, suggestion);


            MessageBox.Show("Thank you for your feedback!", "Feedback Submitted",
MessageBoxButtons.OK, MessageBoxIcon.Information);
```

```
        }
        private int GetCheckBoxRating(GroupBox groupBox)
        {
            foreach (Control control in groupBox.Controls)
            {
                if (control is CheckBox checkBox && checkBox.Checked)
                {
                    if (int.TryParse(checkBox.Tag?.ToString(), out int rating))
                    {
                        return rating;
                    }
                }
            }
            return 0;
        }

        private void SaveFeedbackToDatabase(int easeRating, string technicalIssue, string suggestion)
        {
            using (SqlConnection connection = new SqlConnection(ConnectionString))
            {
                connection.Open();

                string query = "INSERT INTO FeedbackResponses (Rating, TechnicalIssue, Suggestion) "
+"VALUES (@Rating, @TechnicalIssue, @Suggestion)";

                using (SqlCommand command = new SqlCommand(query, connection))
                {
                    command.Parameters.AddWithValue("@Rating", easeRating);
                    command.Parameters.AddWithValue("@TechnicalIssue", technicalIssue);
                    command.Parameters.AddWithValue("@Suggestion", suggestion);

                    command.ExecuteNonQuery();
                }
            }
        }
        private void OpenHomeForm()
        {
            Homepage homePage = new Homepage();
            homePage.Show();
            this.Hide();
        }

        private void button2_Click(object sender, EventArgs e)
        {
            DeleteFeedbackFromDatabase();


            MessageBox.Show("Feedback deleted successfully!", "Feedback Deleted",
MessageBoxButtons.OK, MessageBoxIcon.Information);

            OpenHomeForm();

            this.Close();
        }
        private void DeleteFeedbackFromDatabase()
```

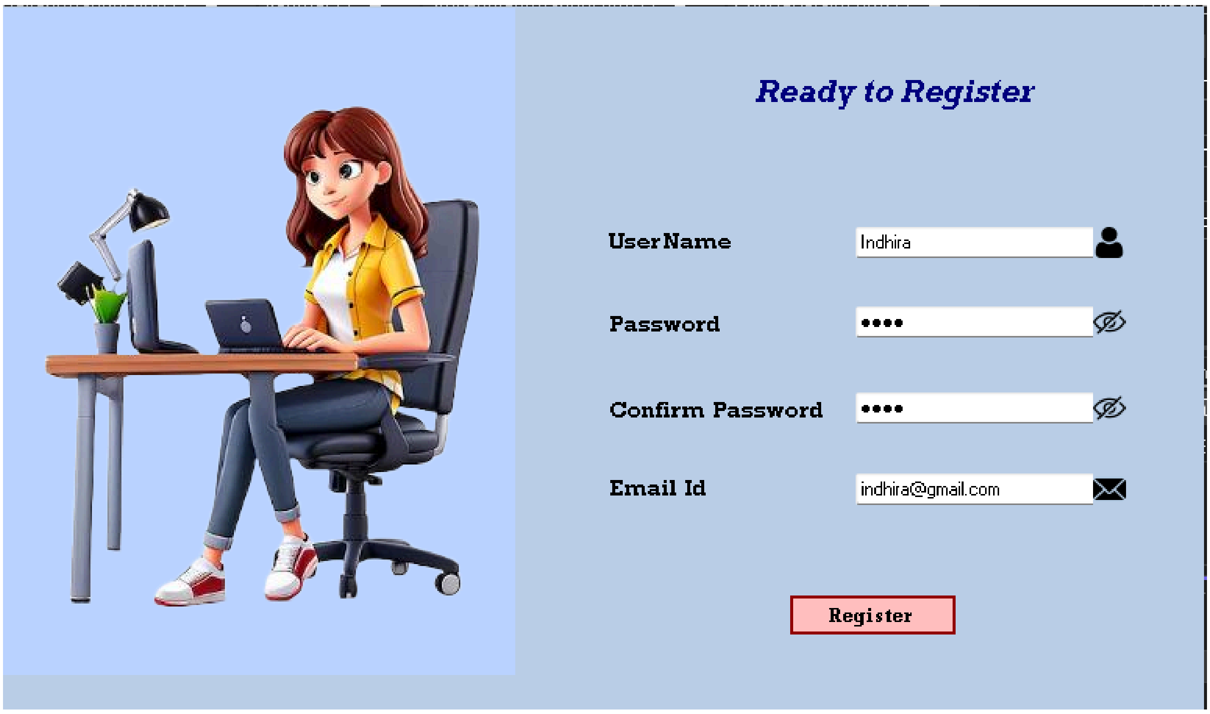
```
        {

            using (SqlConnection connection = new SqlConnection(ConnectionString))
            {
                connection.Open();

                string query = "DELETE FROM FeedbackResponses";

                using (SqlCommand command = new SqlCommand(query, connection))
                {
                    command.ExecuteNonQuery();
                }
            }
        }

        private void button3_Click(object sender, EventArgs e)
        {
            OpenHomeForm();
        }
    }
}
```

SCRREN SHOTS

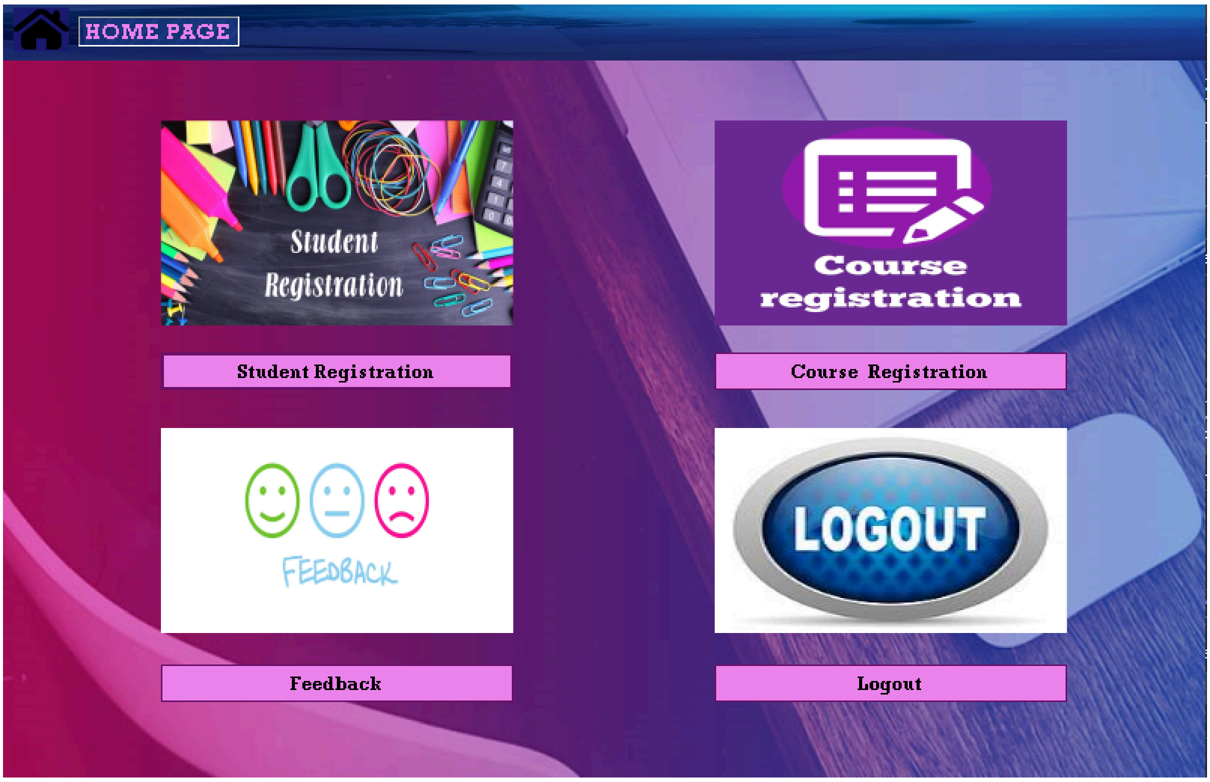
Login Form:



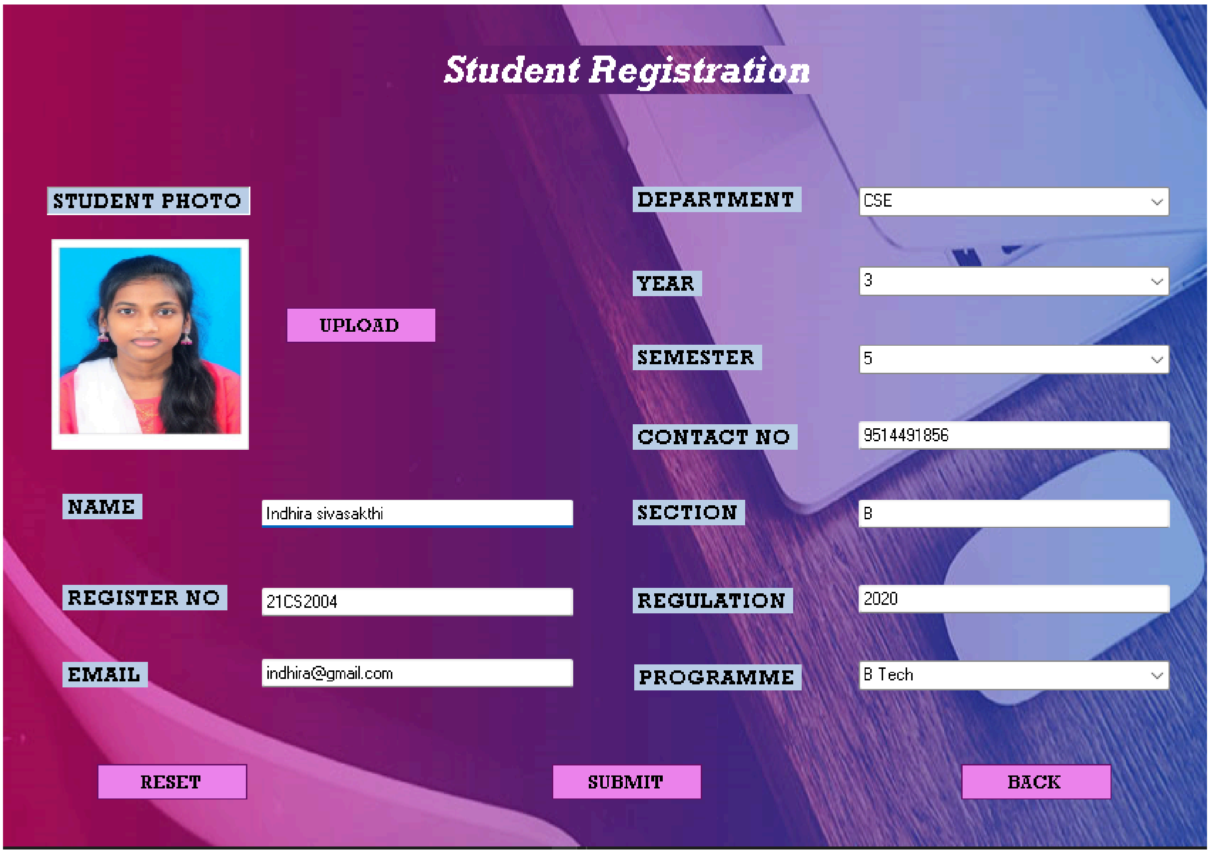
Register Form:

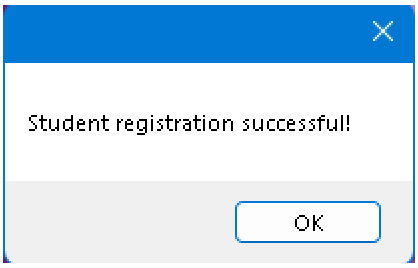


Home Page :

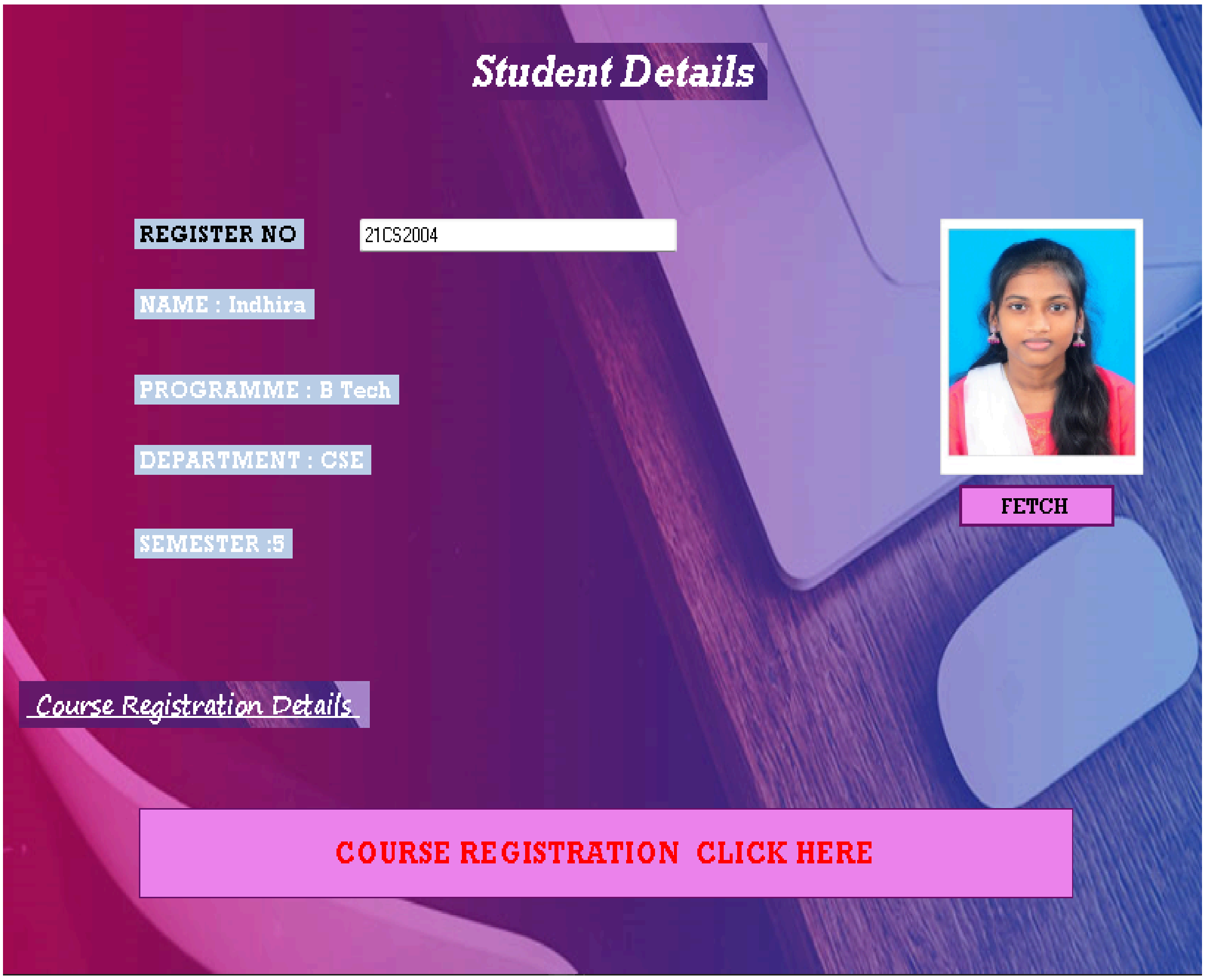


Student Registration Form:



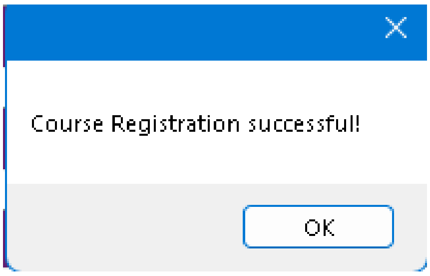


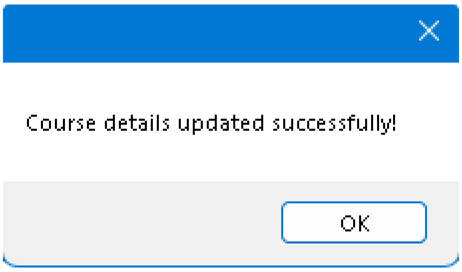
Course Registration Form



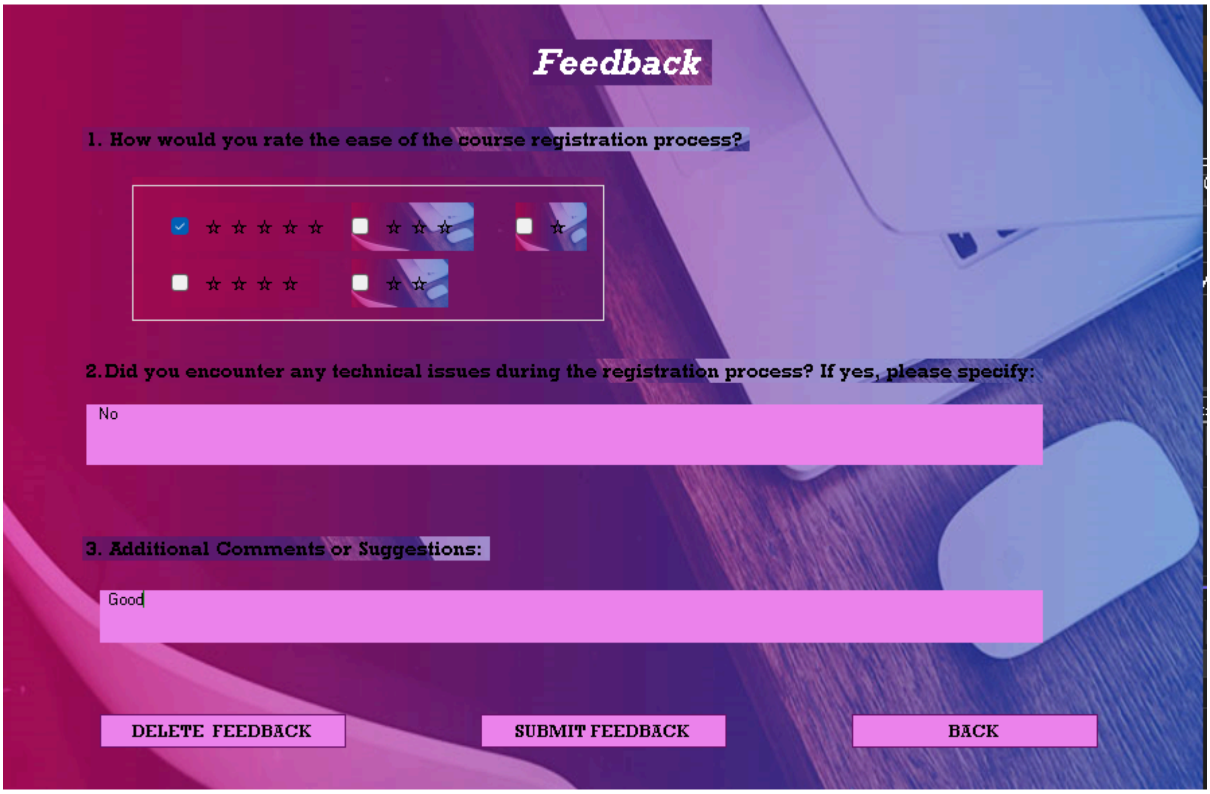
Course Details Form:

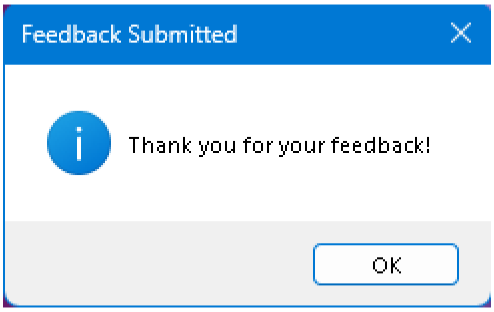
Course Registration Form (continued from the previous page):

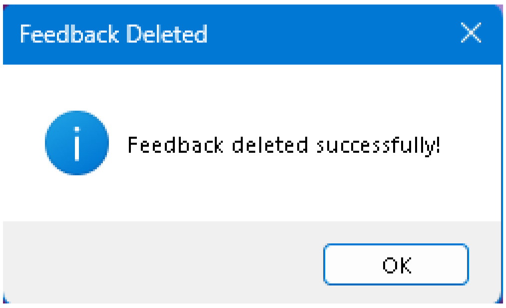




Feedback Form :







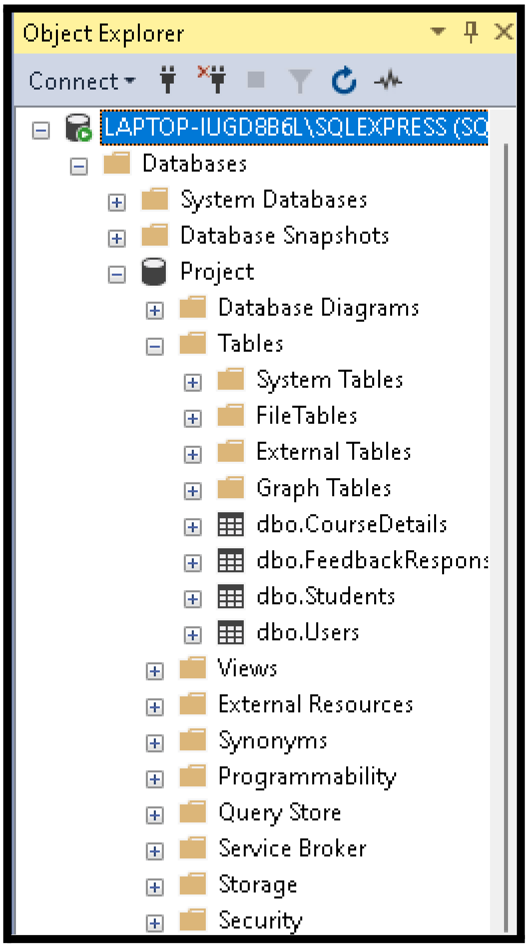
DATABASE

use Project;

```
CREATE TABLE Users (
   UserID INT PRIMARY KEY IDENTITY(1,1),
   Username VARCHAR(50) NOT NULL UNIQUE,
   Password VARCHAR(100) NOT NULL,
   Email VARCHAR(100) NOT NULL UNIQUE
);

select * from Users;

CREATE TABLE Students
(
   StudentID INT PRIMARY KEY IDENTITY(1,1),
   Name VARCHAR(100) NOT NULL,
   RegisterNo VARCHAR(20) NOT NULL,
   Department VARCHAR(50) NOT NULL,
   Semester VARCHAR(10) NOT NULL,
   Year VARCHAR(10) NOT NULL,
   Email VARCHAR(100) NOT NULL,
         ContactNo VARCHAR(10) NOT NULL,
         Section VARCHAR(10) NOT NULL,
         Regulation VARCHAR(10) NOT NULL,
         Programme VARCHAR(10) NOT NULL,
   Photo VARBINARY(MAX) NULL
);

Select * from Students;


CREATE TABLE CourseDetails (
   ID INT PRIMARY KEY IDENTITY(1,1),
   RegisterNo NVARCHAR(50),
   Semester NVARCHAR(50),

   Subject1 NVARCHAR(50),
   SubjectCode1 NVARCHAR(50),
   TheoryLab1 NVARCHAR(50),
   Staff1 NVARCHAR(50),

   Subject2 NVARCHAR(50),
   SubjectCode2 NVARCHAR(50),
   TheoryLab2 NVARCHAR(50),
   Staff2 NVARCHAR(50),

   Subject3 NVARCHAR(50),
   SubjectCode3 NVARCHAR(50),
   TheoryLab3 NVARCHAR(50),
   Staff3 NVARCHAR(50),

   Subject4 NVARCHAR(50),
   SubjectCode4 NVARCHAR(50),
   TheoryLab4 NVARCHAR(50),
   Staff4 NVARCHAR(50),

   Subject5 NVARCHAR(50),
   SubjectCode5 NVARCHAR(50),
```

```
    TheoryLab5 NVARCHAR(50),
    Staff5 NVARCHAR(50),

    Subject6 NVARCHAR(50),
    SubjectCode6 NVARCHAR(50),
    TheoryLab6 NVARCHAR(50),
    Staff6 NVARCHAR(50),

    Subject7 NVARCHAR(50),
    SubjectCode7 NVARCHAR(50),
    TheoryLab7 NVARCHAR(50),
    Staff7 NVARCHAR(50)
);

select * from CourseDetails;

CREATE TABLE FeedbackResponses (
   ID INT PRIMARY KEY IDENTITY(1,1),
    Rating VARCHAR(5),
    TechnicalIssue VARCHAR(30),
    Suggestion VARCHAR(30),
);


select * from FeedbackResponses;
```

PROJECT DATABASE



Users Table:

Results | Messages

	UserID	Username	Password	Email
1	1	Indhira	2121	indhira@gmail.com
2	2	Shwetha	1212	shwetha@gmail.com

Students Table:

Results | Messages

	StudentID	Name	RegisterNo	Department	Semester	Year	Email	ContactNo	Section	Regulation	Programme	Photo
1	1	Indhira	21CS2004	CSE	5	3	indhira@gmail.com	9514491856	B	2020	B Tech	0xFFD8FFE000104A46494600010101012C012C0000FFE113...

CourseDetailsT able:

	ID	RegisterNo	Semester	Subject1	SubjectCode1	TheoryLab1	Staff1	Subject2	SubjectCode2	TheoryLab2	Staff2
1	1	21CS2004	5	Industrial Economics and Management	HS202	Theory	Virappassamy	Platform Technologies	CS215	Theory	Guest Faculty

	Subject3	SubjectCode3	TheoryLab3	Staff3	Subject4	SubjectCode4	TheoryLab4	Staff4	Subject5	SubjectCode5	TheoryLab5	Staff5
1	Computer Networks	CS216	Theory	Sheaba	Automata Theory and Compiler Design	CS217	Theory	Thirumaran	Platform Technologies Laboratory	CS218	Laboratory	Guest Faculty

SubjectCode6	TheoryLab6	Staff6	Subject7	SubjectCode7	TheoryLab7	Staff7
CS219	Laboratory	Sheaba	Essence of Indian Traditional Knowledge	SH203	Theory	Guest Faculty

After update:

Subject7	SubjectCode7	TheoryLab7	Staff7
Essence of Indian Traditional Knowledge	SH203	Mandatory Course	Guest Faculty

FeedbackResponses Table

	ID	Rating	TechnicalIssue	Suggestion
1	1	5	NO	GOOG
2	2	5	NO	GOOD